Appendix

A Detailed description of related work

Methods	Exploration	Affiliation of goal	Individual goal obtaining	Assign credit among agents	Mutual Information	No handcraft knowledge	Interpretable credit assignment
CM3	—	Individual	State	Implicit network	—	✔	✘
CMAE	Entropy	Global	—	Implicit network	—	✔	✘
EITI	VoI	—	—	Counterfact VoI	Transitions	✔	✘
MASER	Entropy	Individual	Observation	Implicit network	—	✔	✘
ALMA	—	Individual	Predefined	Implicit network	—	✘	✘
LAIES	External state	—	—	Implicit network	Transitions	✘	✘
FoX	Formation	—	—	Implicit network	Formation& latent variable	✔	✘
HMASD	Skill	Global& Individual	latent variable	Implicit network	States&skills	✔	✘
Ours	Count within influence scope	Global& Individual	Automatic decomposition	Explicit if-else function	Elements of states&actions	✔	✔

Table 2: Previous work most related to ours

Table 2 summarizes and compares our approach with the most relevant work across seven dimensions. The first dimension is in terms of exploration approach to be token. CMAE (Liu et al., 2021) decides which dimensions to prioritize for exploration by calculating the entropy of each dimension of the state. EITI (Wang et al., 2020) adopts a Value of Interaction (VoI) approach to quantify the influence of one agent's behavior on the expected returns of other agents. MASER adopts the maximum entropy principle to diversify the action choices. LAIES (Liu et al., 2023) specifies the external states of the environment and encourages the agent to explore them. FoX (Jo et al., 2024) investigates the same formation in MARL environments to enhance exploration. HMASD (Yang et al., 2024) enables agents to explore environments more efficiently based on the skills they learn. Different from these approaches, our method's exploration is based on the concept of influence scope introduced in this paper to delimit the exploration space for each agent. In this way, agents obtain count-based intrinsic rewards from the novel states discovered within their respective influence scope.

Table 2 also compares our method with the previous method from the perspective of the goals used in learning. 'Affiliation of goal' in Table 2 indicates whether the goal is designed for the global (team) or for the individual, and 'individual goal obtaining' indicates how the representation of the individual goal is obtained. Our approach includes both global and individual goals. A global goal is a state from the environment, while an individual goal is automatically decomposed from this global goal based on the influence scope of the individual agent.

As far as credit assignment is concerned, most previous approaches involving network policies have used a network to implicitly assign credit among agents through backpropagation of the gradient. EITI utilizes its proposed VoI to exclude the contribution of other agents through its counterfactual form when evaluating individual behaviors. The credit assignment in our method is a explicit if-else process. Our method determines whether the current actions of the agents impact a segment of the goal. If they do not, the reward for that segment will not be assigned to the agents.

Moreover, when it comes to the computation of mutual information, our approach considers the mutual information of each particular action of the agent on each dimension of the state value. This differs from calculating the mutual information of information or skill with respect to a hidden variable, and from calculating the mutual information between transitions. Our approach does not involve handcrafted knowledge, and the credit assignment is interpretable due to its if-else process.

B Detailed description for the loss and network structure

We follow IPPO loss De Witt et al. (2020) to update $\{\pi_i^e\}_{i\in\mathcal{I}}$ and $\{\pi_i\}_{i\in\mathcal{I}}$. The exploration policies $\{\pi_i^e\}_{i\in\mathcal{I}}$ are trained with Equ.(10) and the environment reward. Specifically, for agent i's exploration, the TD error is calculated with $\eta_{i+}^t = r_{i+} + \beta_2 r + \gamma V_i^{\phi_e}(\tau_i^{t+1}) - V_i^{\phi_e}(\tau_i^t)$, where r_{i+} is exploration reward from Equ. (10), r is the environmental reward, β_2 is the factor, γ is the discount factor in Dec-POMDP, $V_i^{\phi_e}$ is the value estimation function parameterized by ϕ_e, $\tau_i^{t+1}, \tau_i^1 \in \mathcal{T}_i$ represent the local trajectories at $t+1$ and t. The advantage of its exploration policy π_i^e is estimated by:

$$A_{i+}^t = \sum_{l=0}^{h} (\gamma\lambda_{gae})^l \eta_{i+}^{t+l}$$

where h and λ_{gae} are the total step number and the Generalized Advantage Estimation (GAE) factor (Schulman et al., 2016). The loss to train value function is:

$$\mathcal{L}_{i+}(\phi_e) = \mathbb{E}_{\tau_i^t}\left[\min\left\{\left(V_i^{\phi_e}(\tau_i^t) - \hat{V}_i^{\phi_e}(\tau_i^t)\right)^2, \left(V_i^{\phi_e^{old}}(\tau_i^t) + clip(V_i^{\phi_e}(\tau_i^t) - V_i^{\phi_e^{old}}(\tau_i^t), -c, +c) - \hat{V}_i^{\phi_e}(\tau_i^t)\right)^2\right\}\right]$$

where $\hat{V}_i^{\phi_e}(\tau_i^t) = A_{i+}^t + V_i^{\phi_e}(\tau_i^t)$, ϕ_e^{old} are old parameters before the update, $clip(V_i^{\phi_e}(\tau_i^t) - V_i^{\phi_e^{old}}(\tau_i^t), -c, +c)$ restricts the update of the value function to within the trust region $(-c, +c)$ and c is hyperparameter. The policy loss for agent i is:

$$\mathcal{L}_{i+}(\theta_e) = \mathbb{E}_{\tau_i^t, a_i^t}\left[\min\left\{\frac{\pi_i^{\theta_e}(a_i^t|\tau_i^t)}{\pi_i^{\theta_e^{old}}(a_i^t|\tau_i^t)} A_{i+}^t, clip(\frac{\pi_i^{\theta_e}(a_i^t|\tau_i^t)}{\pi_i^{\theta_e^{old}}(a_i^t|\tau_i^t)}, 1-c, 1+c) A_{i+}^t\right\}\right]$$

where $\pi_i^{\theta_e}$ represents the π_i^e parameterized by θ_e and $\pi_i^{\theta_e^{old}}$ is the old policy before the update. The overall learning loss is:

$$\mathcal{L}_e = \sum_{i=1}^{|\mathcal{I}|}\left(\mathcal{L}_{i+}(\theta_e) + \lambda_{critic}\mathcal{L}_{i+}(\phi_e) + \lambda_{entropy}\mathcal{H}(\pi_i^{\theta_e})\right)$$

where $\mathcal{H}(\pi_i^{\theta_e})$ denotes the entropy of policy $\pi_i^{\theta_e}$, λ_{critic} and $\lambda_{entropy}$ are hyperparameter factors.

The goal-conditioned policies $\{\pi_i\}_{i\in\mathcal{I}}$ are trained with Equ. (7) and the environment reward. The loss function of $\{\pi_i\}_{i\in\mathcal{I}}$ is same with that of $\{\pi_i^e\}_{i\in\mathcal{I}}$ but replace all the input τ_i^t with (τ_i^t, g_i). Specifically, for agent i's goal-conditioned policy, the TD error is calculated with $\eta_i^t = r_i + \alpha_2 r + \gamma V_i^{\phi}(\tau_i^{t+1}, g_i) - V_i^{\phi_e}(\tau_i^t, g_i)$, where r_i is goal-conditioned reward from Equ. (7), r is the environmental reward, α_2 is the factor, V_i^{ϕ} is the value estimation function parameterized by ϕ. The advantage of its goal-conditioned policy π_i is estimated by:

$$A_i^t = \sum_{l=0}^{h} (\gamma\lambda_{gae})^l \eta_i^{t+l}$$

The loss to train value function is:

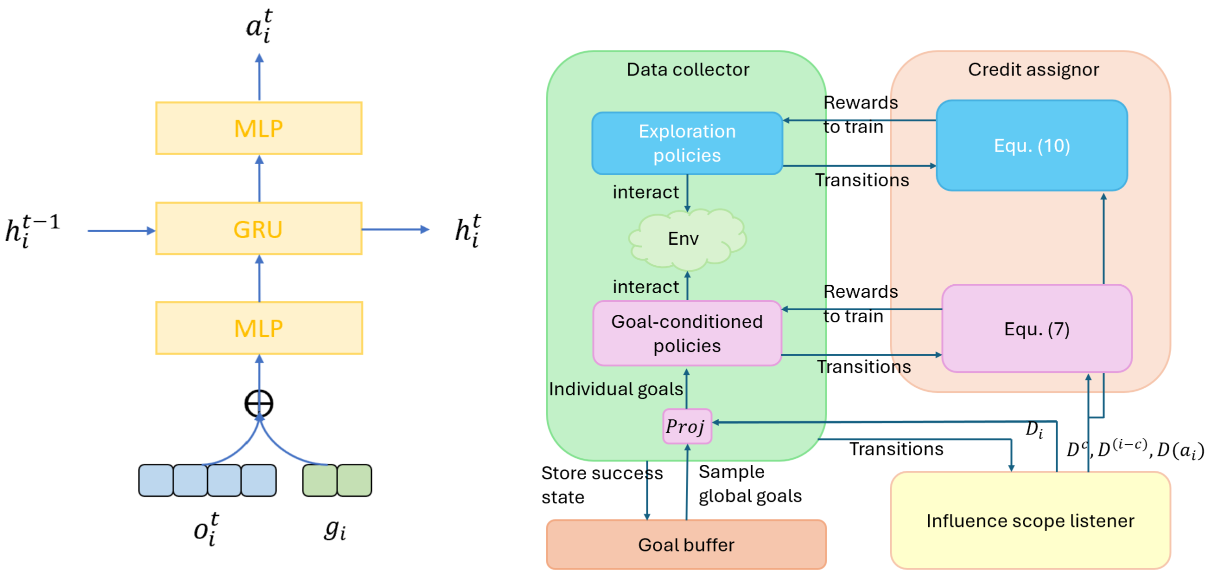
$$\mathcal{L}_i(\phi) = \mathbb{E}_{\tau_i^t}\left[\min\left\{\left(V_i^{\phi}(\tau_i^t, g_i) - \hat{V}_i^{\phi}(\tau_i^t, g_i)\right)^2, \left(V_i^{\phi^{old}}(\tau_i^t, , g_i) + clip(V_i^{\phi}(\tau_i^t, g_i) - V_i^{\phi_e^{old}}(\tau_i^t, g_i), -c, +c) - \hat{V}_i^{\phi_e}(\tau_i^t, g_i)\right)^2\right\}\right]$$

where $\hat{V}_i^{\phi}(\tau_i^t, g_i) = A_i^t + V_i^{\phi}(\tau_i^t, g_i)$, ϕ^{old} are old parameters before the update. The policy loss for agent i is:

$$\mathcal{L}_i(\theta) = \mathbb{E}_{\tau_i^t, a_i^t}\left[\min\left\{\frac{\pi_i^{\theta}(a_i^t|\tau_i^t, g_i)}{\pi_i^{\theta^{old}}(a_i^t|\tau_i^t, g_i)} A_i^t, clip(\frac{\pi_i^{\theta}(a_i^t|\tau_i^t, g_i)}{\pi_i^{\theta^{old}}(a_i^t|\tau_i^t, g_i)}, 1-c, 1+c) A_i^t\right\}\right]$$

where π_i^{θ} represents the π_i parameterized by θ and $\pi_i^{\theta^{old}}$ is the old policy before the update. The overall learning loss is:

$$\mathcal{L} = \sum_{i=1}^{|\mathcal{I}|}\left(\mathcal{L}_i(\theta) + \lambda_{critic}\mathcal{L}_i(\phi) + \lambda_{entropy}\mathcal{H}(\pi_i^{\theta})\right)$$



(a) Goal-conditioned policy network

(b) ISA framework

Figure 4: Network structure and framework of ISA.

where $\mathcal{H}(\pi_i^\theta)$ denotes the entropy of policy π_i^θ. The network structure for an individual goal-conditioned policy is the same with IPPO but the input vector connection of the observation and goal vectors as shown in Fig. 4(a). At each time step t, the observation o_i^t and goal g_i are first connected together. The action a_i^t is then obtained through the Multilayer Perceptron (MLP) (Cybenko, 1989), followed by the Gated Recurrent Units (GRU) (Chung et al., 2014), and another MLP. The history trajectory is encoded by GRU as h_i^{t-1} at last time step, which is encoded along with o_i^t and g_i as h_i^t for the next time step.

Fig. 4(b) shows the four modules in ISA framework:

- **Influence scope listener.** Applying on interaction trajectories, the influence scope listener is responsible for calculating the influence scopes $D(a_i)$ and D_i for each $i \in \mathcal{I}$ and each $a_i \in A_i$ according to Definitions 1 $\sim$ 3, which underlays the exploration and policy learning in our work.

- **Data collector.** The data collector is responsible for collecting data from environments, which includes exploration policies $\{\pi_i^e\}_{i\in\mathcal{I}}$ and goal-conditioned policies $\{\pi_i\}_{i\in\mathcal{I}}$. These two set of policies are trained separately with different rewards. Exploration policies are trained to more efficiently discover the success state of the Dec-POMDP. The goal-conditioned policies are trained to make optimal joint decisions to achieve success states.

- **Goal buffer.** The goal buffer is used to store success states during interaction for training goal-conditioned policies.

- **Credit assignor.** The credit assignor is in charge of providing quantifiable feedback from state change and assigning credits among agents, which consists of Equ. (10) for training exploration policies $\{\pi_i^e\}_{i\in\mathcal{I}}$ and Equ. (7) for training goal-conditioned policies $\{\pi_i\}_{i\in\mathcal{I}}$.

C Detailed descriptions on experiments

C.1 Environments

SMAC is a real-time strategy game, which is widely used as an MARL benchmark. SMAC has different reward setting as shown in Table 3. In dense reward setting, agents intensively receive immediate reward for their actions, whereas in sparse and super sparse setting the rewards are delayed where the agents is rewarded only when rare events occur. In our experiments, we take the most challenging super sparse reward setting to validate the independence of our method from handcrafted rewards in sparse-reward tasks.

Event	Dense Reward	Sparse Reward	Super Sparse Reward
Enemy hit-point h_e	-h_e	0	0
Ally hit-point h_a	$+h_a$	0	0
An enemy death	+10	+10	0
An ally death	-5	-5	0
Win	+200	+200	+1
Loss	0	0	-1

Table 3: The reward setting for SMAC domain.

Although SMAC provides different scenarios for agents to learn various cooperative skills, the success of most of its scenarios depends on the actual D^c (i.e., enemy health) in the environment rather than on the actual $D^{(i-c)}$(i.e., coordinates of individual agents, etc.). Therefore, we also consider Navigation, Shooting and Unlock tasks on MPE. The Navigation task requires agents to learn to occupy different landmarks. In this task, the state information include positions of all agents and landmarks. This is a typical task where there is no obvious inherent joint influence among agents (i.e., $D^c = \emptyset$). We use this task to validate the performance of ISA in environment where the environment rewards depend on D^{i-c}. The Shooting tasks requires agents to shoot a target enough times and move towards specific positions. In this task, the state attributes include the times of the target having been shoted (jointly influenced by all agents) and positions of individual agents (influenced by individual agents). We use this task to validate ISA where the rewards depend on both inherent D^c and D^{i-c} in the environment. In the Unlock task, an individual agent, even if it does not take any actions, will still receive a team reward for opening a lock by another agent. We use this task to verify the resistance of ISA agent to interference of team reward caused by actions of other agents. The state of SMAC is continuous, while the state of MPE is discrete. Their specific environmental information is shown in Table 4.

	number of states	number of actions	number of agents	dimension of state
3m	$+\infty$	9	3	46
2s_vs_1sc	$+\infty$	7	2	25
8m	$+\infty$	14	8	166
Navigation	20^{14}	5	7	14
Shooting	$20^6 \times 50$	6	3	7
Unlock	20^{14}	5	7	14

Table 4: Environmental information for tasks in SMAC and MPE

C.2 Baselines

QMIX, COMA, and IPPO are 3 classical MARL algorithms. QMIX using a mixing network to assign credit among agents. COMA trains a counterfactual value function for credit assignment. IPPO directly uses team reward as individual rewards. CMAE is an exploration method designed for sparse-reward multi-agent tasks. MASER, like our method, employs a distance-based intrinsic reward measure for individual goal achievement, but the representation of individual goals is encoded from observations. FoX is an state-of-art exploration MARL method in terms of performance in tasks with few immediate rewards. HMASD explores and learns the tasks with skills, which is the state-of-art method in terms of performance in tasks with super sparse setting of SMAC. For the implement of MASER, FoX, and HMASD, we use the code released by the authors. For the implement of classical MARL algorithms QMIX, COMA, and IPPO, we employ a widely code framework released by (Hu et al., 2021). Because the code provided by authors of CMAE only work for 2 agents environments and the do not provide code for SMAC domain, we use the performance data reported in their paper.

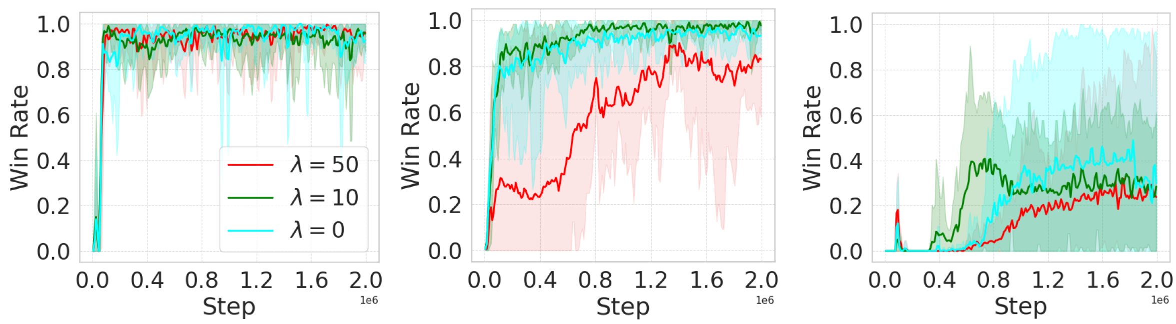
C.3 Hyperparameters

We show the detailed hyperparameter settings for ISA in Table 5. During the development of ISA, we tried different hyperparameters. As elaborated in Remark 1, too large or too small values for threshold δ are not

	3m	2s_vs_1sc	8m	Navigation	Shooting	Unlock	Description
δ	0.3	0.3	0.3	0.3	0.3	0.3	Threshold in Equ. (4)
λ	50	50	10	0	0	0	Scale factor for Hamming distance
α_1, β_1	0	0	0	0.2	0.2	0.2	Scale factors in Equ. (7) and (10)
α_2, β_2	0	0	10	1	1	1	Scale factors for environment reward
N	2,000	2,000	10,000	2,000	2,000	2,000	Number of transitions in Alg. 1
L	1	1	1	1	1	1	Goal buffer Length in Alg. 1
w_{bin}	0.01	0.05	0.01	0.01	0.01	0.01	Width of equal binning to discretize Δs^k
l_{hidden}	64	64	64	12	12	12	Size of hidden layer
l_{buffer}	64	64	64	32	32	32	Size of episode buffer for policy training
$n_{running}$	8	8	8	8	8	8	Number of processes for data collection
$n_{training}$	64	64	64	32	32	32	Batch size for policy training
lr	0.0005	0.0005	0.0002	0.0005	0.0005	0.0005	Learning rate
λ_{critic}	0.5	0.5	0.5	0.5	0.5	0.5	Scale factor for updating critic
$\lambda_{entropy}$	0.001	0.001	0.001	0.001	0.001	0.001	Scale factor for entropy regularization
λ_{gae}	0.95	0.95	0.95	0.95	0.95	0.95	GAE factor
c	0.2	0.2	0.2	0.2	0.2	0.2	Clipping value to restrict the update

Table 5: Detailed hyperparameter setting to reproduce the performance of the ISA in Figure 1

inappropriate to capture the inherent influence scope of agent in the environment. In the development of ISA, we tried values of δ between 0 and 0.6. The results show that the inherent influence scope of agents can be accurately captured when $\delta \in [0.15, 0.45]$ in the 3m task. We finally chose $\delta = 0.3$ and applied it to all other tasks. The scale factor for Hamming distance is used to scale the intrinsic reward whether or not each value of attribute on current state achieving the value in the goals. During the development, we tried the value of λ from $\{0, 10, 50\}$ in SMAC domain. For factors α_2, β_2 that scale the environmental reward, we tried their value from $\{0, 10, 50\}$ in the 8m task. In addition, because there are more agents and more individual actions in 8m, we took a larger $N = 10000$ in this task to collect more data to evaluate the mutual information. Besides, we tried larger $w_{bin} = 0.05$ on 2s_vs_1sc to ignore the minor changes in the state's attributes, and tried smaller $lr = 0.0002$ on 8m to stabilize the learning. Apart from these mentioned tries, the other hyperparameters always adopt the values in Table 5 during the development of ISA.



(a) 2s_vs_1sc (b) 3m (c) 8m

Figure 5: Performance of ISA with different λ.

We show the performance of ISA with different λ in Fig. 5. The experiment in Figure 5 changes the value of λ while the values of the other hyperparameters remain unchanged and follow the values in Table 5.

C.4 Statistical tests on improvement

We also conduct statistical tests using the Wilcoxon signed-rank test to compare the performance improvements brought by ISA against the state-of-the-art performance across the 6 tasks. The data used for the statistical test come from 25 sets of final performance evaluation from 5 different random seeds after convergence. The p-values is shown in Table 6. The p-values in all tasks are sufficiently small (less than 0.05) except for the 3m task. This suggests that we can reject the hypothesis that the final performance differences between ISA and the state-of-the-art methods in these tasks are due to chance.

	3m	2s_vs_1sc	8m	Navigation	Shooting	Unlock
p-value	7.05×10^{-2}	2.98×10^{-8}	6.51×10^{-5}	1.73×10^{-2}	5.96×10^{-8}	1.19×10^{-7}

Table 6: Significance of ISA improvement on state-of-the-art in final performance with Wilcoxon signed-rank tests

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